

Title

Assessment of neuromuscular function after different strength training protocols using
tensiomyography

Running head

Evaluation of tensiomyography after strength exercise

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ABSTRACT

The purpose of the study was to analyse tensiomyography (TMG) sensitivity to changes in muscle force and neuromuscular function of the muscle rectus femoris (RF) using TMG muscle properties after five different lower-limb strength training protocols (MS = Multiple Sets; DS = Drop Sets; EO = Eccentric Overload; FW = Flywheel; PL = Plyometrics). After baseline measurements, 14 male strength trained athletes completed one squat training protocol per week over a five-week period in a randomized controlled order. Maximal voluntary isometric contraction (MVIC), TMG measurements of maximal radial displacement of the muscle belly (Dm), contraction time between 10 and 90% of Dm (Tc) as well as mean muscle contraction velocities from the beginning until 10% (V_{10}), and 90% of Dm (V_{90}) were analysed up to 0.5 hours (post-train), 24 hours (post-24) and 48 hours (post-48) after the training interventions. Significant ANOVA main effects for measurement points were found for all TMG contractile properties and MVIC ($p < 0.01$). Dm and V_{10} post-train values were significantly lower after protocols DS and FW compared to protocol PL ($p = 0.032$ and 0.012 , respectively). Dm, V_{10} , and V_{90} decrements correlated significantly to the decreases in MVIC ($r = 0.64$ - 0.67 ; $p < 0.05$). Some TMG muscle properties are sensitive to changes in muscle force and different lower-limb strength training protocols lead to changes in neuromuscular function of RF. In addition, those protocols involving high and eccentric load, and a high total time under tension may induce higher changes in TMG muscle properties.

Key words: muscle contractile properties, force, strength exercises

INTRODUCTION

Effective strength training programs should include intensive and fatigue-inducing exercise sessions and their acute physiological responses depend on multiple strength exercise determinants such as load intensity, number of repetitions, rest period between sets, type of muscle action, time under tension, and the mechanics of training machines (17,34). The knowledge of the effects of different strength training protocols on fatigue and their respective recovery demands is essential for a well-planned strength training. While too low strength training loads have no impact on physiological systems, very high training loads may increase the likelihood of injury or overtraining symptoms (5). Thus, the acute effects of different types of strength training programs have been investigated in relation to performance measurements, physiological markers and psychometrical instruments, such as countermovement jump (7), maximal voluntary isometric contraction (7,15,20), serum creatine kinase (7,15,20), blood lactate concentration (7,38,39), ratings of perceived exertion (39), and delayed onset of muscle soreness (7,18,21).

In this context, tensiomyography (TMG) may have an additional advantage for strength training monitoring. TMG measures can be carried out quickly, without producing additional fatigue and does not depend on voluntary motivation. It allows a non-invasive neuromuscular function assessment, through different specific skeletal muscle contractile properties, including muscle belly radial stiffness, time and speed of contraction, and muscle fibre distribution (8,12,13,15,25). The intraclass correlation coefficient scores for different TMG muscle contractile properties ranged from 0.85 to 0.95 in previous studies and reflect a sufficient reproducibility (25,30). Acute influence from different types of exercise, such as ultra-endurance triathlon (13), high-load and high-volume resistance exercise (12), and eccentric exercise (15), on TMG muscle properties has been investigated. Loss in contractile capacity, by means of an increase in muscle contraction time of the muscle biceps femoris

measured with TMG, of nineteen males was observed after an ultra-endurance triathlon (13). The impairment of velocity of contraction, might be caused by neural mechanisms and altered intracellular Ca^{2+} regulation (12,13).

However, as far as we know, no study has compared the acute effects of a wide range of strength training protocols used in the practical field on neuromuscular function using TMG muscle properties.

Therefore, the purpose of the present study was to analyse the TMG sensitivity to changes in muscle force and neuromuscular function of the muscle rectus femoris (RF) using TMG muscle properties, after five different lower-body strength training protocols in male strength trained athletes. We hypothesized that TMG is able to detect changes in muscle force and that different strength training protocols exert different impacts on TMG muscle contractile properties.

METHODS

Experimental Approach to the Problem

TMG contractile properties were assessed before and after five different lower body strength training protocols in a randomized cross-over design with multiple repeated measures (Figure 1). All participants attended a familiarization session to introduce the testing and training procedures to minimize any learning effect. Average baseline values were collected on two occasions interspaced by one week including measures of body composition, TMG contractile properties, one repetition maximum for parallel squat (1RM) and maximal voluntary isometric contraction (MVIC) for the same exercise. The following main experimental period consisted of five different training protocols separated by six days in between. Training protocols were randomly assigned for each participant and performed once per week within

1.5 - 2 hours at the same time of the day throughout the study. It was told to the subjects not to exercise from the day before training until the time of 48 hours post-exercise, and to consume their last meal (caffeine-free) at least 2 hours before training and testing. TMG followed by MVIC measurements were conducted up to 0.5 hours after the end of training (post-train) as well after 24 hours (post-24) and 48 hours (post-48) and compared to baseline measures. MVIC measurements were conducted only at baseline and post-24. In order to evaluate metabolic demands and rate of perceived exertion (RPE) from each protocol, blood lactate concentrations (La) and CR-10 RPE scale were measured immediately after the training interventions.

Subjects

Fourteen healthy male (age: 23.0 ± 1.9 years; body mass: 76.6 ± 7.8 kg; height: 179.4 ± 6.8 cm), experienced in strength training for at least two years with a minimum of two training units per week and free of orthopaedic disease participated in the study. As inclusion criterion, the subjects had to be able to achieve at least 100% of their body weight in one-repetition maximum in parallel squat. Subjects' parallel squat performance data were 113.2 ± 16.3 kg and 1.5 ± 0.2 kg.BW⁻¹. Subjects were informed about all details of the experimental procedures and the associated risks and discomforts. All participants gave their written consent to participate in the study and were free to withdraw from the study at any time. The experimental protocol followed the world medical association's declaration of Helsinki on research with humans and was approved by the local Ethics Committee of the Ruhr-University Bochum.

Training Protocols

Regarding the exercise prescribed, the squat was chosen as basic movement because of its similar biomechanical and neuromuscular demands with several athletic movements. Moreover, squat is one of the most popular exercises in strength and conditioning settings,

considered an outstanding evaluation of lower-body strength (29). All training sessions were preceded by a 5 min standardized warm-up (i.e., slow jogging) and core stability exercises (i.e., plank, side plank). Table 1 shows the five different squat variations.

Multiple Sets (MS): A smith rack machine with a guided barbell was used for training (TechnoGym Multipower, Italy). The protocol consisted of 4 sets of 6RM (i.e., 85% 1RM) parallel squats (knees are flexed until the inguinal fold is in a straight horizontal line with the top of the knee musculature), intended explosive during the concentric phase and 2 seconds in eccentric phase, approximately 72 seconds of time under tension (TUT), and 3 minutes rest between sets (10). A laser imager and an acoustic stimulus were used to standardize the ROM of approximately 110-120°.

Drop Sets (DS): Subjects performed DS with the same barbell machine and ROM as described for MS. 1 set of 6RM (i.e., 85% 1RM), 4 seconds in eccentric and 2 seconds in concentric phase respectively, and approximately 130-150 seconds TUT was conducted (31). Immediately after the first set, the load was reduced for the next three sets (70%, 55% and 40% 1RM, respectively), so that the subjects continued to train until concentric failure for each load, which was defined as the point when the muscles involved can no longer produce force enough to sustain the given load (39).

Eccentric Overload (EO): This protocol combined concentric with enhanced eccentric muscle actions (39) with the same barbell machine and ROM as the two protocols described before. 4 sets of 6 repetitions at a load of 70% concentric and 100% eccentric of their individual 1RM, 3 minutes rest between sets, were performed during approximately 4 seconds each repetition (i.e., 2 seconds eccentric, 1 second isometric, and intended explosive in concentric phase), and approximately 96 seconds TUT. Weight changes were organized by two helpers during the upright and lower position.

Flywheel (FW): A YoYo squat flywheel machine was used for training (YoYo Technology, Stockholm, Sweden). Subjects performed 4 sets of 6 maximal repetitions, approximately 96 seconds TUT, 3 minutes rest between sets. Besides 6 intended maximal repetitions, 2 previous repetitions were selected for initial movement acceleration. The squat movement was executed with a ROM of about 95-105°, starting the concentric action at approximately 60-70° until about 165° of internal knee angle, carefully controlled by an experienced supervisor (21). Subjects were asked to perform each repetition with a maximum effort, accelerating the wheel in the concentric action and upon completion, decelerating the wheel by means of an eccentric action.

Plyometrics (PL): Subjects performed 4 sets of 15 drop jumps from a 60 cm-jump box, with 5 seconds rest between repetitions, and 3 minutes rest between sets (35). The study participants were asked to land until the knees are flexed of about 90° followed by a simultaneous explosive knee extension and arms swing for maximum vertical jump height.

Figure 1

Table 1

Measurements

One Repetition Maximum (1RM): The hypothetical 1RM for each participant in a smith rack machine (TechnoGym Multipower, Italy) was assessed by the formula proposed by Brzycki (4). Subjects were instructed to position into a shoulder bride stand and the barbell was placed on the trapezius muscle and posterior deltoid muscle. In the parallel squat, the knees are

flexed until the inguinal fold is in a straight horizontal line with the top of the knee musculature. A laser imager and an acoustic stimulus served to standardize the ROM of approximately 105-110°. Subjects started with two warm-up sets consisting of five repetitions with an intensity of 50% of the individual body weight with two minutes pause. After that, a work set including five repetitions with an intensity of 80 to 85% of the individual body weight was performed. Finally, after five minutes, the test supervisor asked to increase the weight for estimating the 1RM. The test was stopped when subjects were unable to raise the barbell with a proper technique or without the help of the supervisor. If subjects exceeded the limit of ten repetitions, the supervisor stopped the test and the intensity was increased. The test ended when subjects achieved five to ten maximum repetitions and the 1RM was estimated in kg.

Maximal Voluntary Isometric Contraction (MVIC): MVIC was measured in a half squat isometric exercise using a Multitrainer 7812-000 (Kettler Profiline, Germany) and analogous user software (DigiMax Version 7.X). The subjects were directed to position under the shoulder upholstery into a shoulder bride stand. Subsequently the subject was set up into a testing position up to a knee-joint angle of 90° using a custom made goniometer. Without moving explosively, but low rate of force development they were asked to produce a maximal voluntary isometric contraction over a 3 second time interval, as recommended by Blazeovich et al. (2). All subjects performed two MVICs with 2 min rest in between and the mean of both attempts was recorded.

Tensiomyography (TMG): TMG measurements were conducted using a specific electrical stimulator (TMG-S2), the TMG-OK 3.0 software, as well as a displacement sensor (Figure 2) tip with a prefixed tension of 0.17 N m^{-1} which was positioned rectangularly to the muscle belly (TMG-BMC, Ljubljana, Slovenia). The measuring point for each muscle was carefully determined as a point of maximal muscle belly displacement during voluntary knee extension.

The contractile properties assessed were the maximal radial displacement of the muscle belly (Dm), contraction time between 10 and 90% Dm (Tc), mean velocity until 10% Dm (V_{10}) and mean velocity until 90% Dm (V_{90}). These constitute the main parameters in this trial due to their higher precision and sensitivity (11,25,30). The contractile properties of the left and right RF were analysed during a twitch contraction evoked by individual maximal electrical stimulation over the muscle belly of 1 ms duration. Maximal electrical stimulation and maximal muscle belly displacement were found by progressively increasing the electric current by 20 mA for each stimulation, each time separated by 10-second intervals to minimize the effects of fatigue and potentiation (25). The maximal response was usually achieved around 100 mA. The average value from the both legs of two maximal twitches was used for further analyses. The RF was assessed in a supine position and a knee joint angle of 120° was kept by using supporting pads. The electrodes (5 x 5 cm) were placed five cm distally and five cm proximally to the sensor. The positions of electrodes and sensor were marked and kept constant during the complete experimental period. All the measurement procedures were accomplished according Rey et al. (25).

Figure 2

In a preliminary study (30), reproducibility of TMG muscle properties was assessed with 20 sport students for rectus femoris, biceps femoris and gastrocnemius lateralis. All the muscles analysed, the intraclass correlation coefficient (ICC) scores ranged from 0.85 to 0.95 and reflect a sufficient reproducibility. RF was selected due to the highest ICC scores obtained against biceps femoris and gastrocnemius lateralis. The ICC scores for the muscle RF for Dm, V_{10} , V_{90} , and Tc were 0.92, 0.91, 0.90, and 0.93 respectively. ANOVA systematic error for these parameters showed no difference between trials ($p > 0.05$), whereas the standard error of

the mean (SEM) represented 11.4, 11.1, 12.9, and 7.1% for the same variables.

RPE and Blood Lactate (La): Subjects were asked to rate the CR-10 RPE scale immediately after the last repetition of each protocol (3). A rating of 0 is equivalent to rest (i.e., no effort) and a rating of 10 was quantified as maximal effort. Immediately after the RPE rating, earlobe capillary blood samples (20 µl) were collected into a capillary tube. The samples were frozen at -20°C and analysed by EBIOplus kit (Eppendorf AG, Germany).

Body composition: The skinfolds thickness from the both thighs was measured (Harpender Skinfold Caliper, UK) exactly where the TMG sensor was positioned during baseline and the average value calculated, to verify possible influences from subcutaneous fat on TMG measurements.

Statistical analyses

Data are presented as the mean $\pm$ standard deviation (SD). These data were analysed using the Statistical Package for the Social Sciences 18.0 Software (SPSS Inc., USA). The Kolmogorov-Smirnov test was used to check the normality of the data distribution. A two-way repeated measures ANOVA (5 protocols x 4 measurement times) was used to determine the effects of different strength training protocols over time on muscle properties Dm, Tc, V₁₀ and V₉₀. The effects of the different strength training protocols on muscle force over time were determined using a two-way repeated measures ANOVA [5 protocols x 2 measurement times (baseline and post-test)]. The effects of the different strength training protocols on La and RPE at post-train were determined using a one-way repeated measures ANOVA. The post-hoc analyses included paired t-tests with Bonferroni corrections. After the calculation of the mean values of changes in TMG parameters and MVIC from the five protocols from baseline to post-train, Pearson correlation coefficient was used to establish the relationship between these changes and the relationship between TMG parameters and skinfolds thickness of the thighs as well. Statistical significance was set at $p < 0.05$.

RESULTS

Post exercise La and RPE values were significantly different between the protocols ($F_{[2,28]} = 56.16$, $p < 0.001$ and $F_{[2,28]} = 58.71$, $p < 0.001$ respectively) and are presented in Table 2. The Bonferroni correction showed that the lowest La concentration was found after PL (3.0 ± 1.6 mmol.L⁻¹, $p < 0.001$) whereas FW was the most glycolytic stressful protocol (12.0 ± 1.8 mmol.L⁻¹, $p < 0.01$). Bonferroni corrected paired t-tests showed that RPE demonstrated a significant lower perceived effort after PL (5.2 ± 1.6 , $p < 0.05$) compared to all other protocols. The RPE values after FW (9.9 ± 0.3) were also significantly higher ($p < 0.05$) compared to MS and DS respectively (8.6 ± 1.0 ; 9.0 ± 0.9).

Table 2

Significant ANOVA main effects for measurement points were found for Dm ($F_{[1,17]} = 16.83$, $p < 0.001$), Tc ($F_{[2,21]} = 20.07$, $p < 0.001$), V_{10} ($F_{[1,17]} = 12.63$, $p < 0.01$), V_{90} ($F_{[1,16]} = 11.08$, $p < 0.01$). With the exception of PL, in all other training protocols, paired t-tests with Bonferroni correction showed a decrease in Dm, Tc, V_{10} and V_{90} from baseline to after training (post-train). Dm, V_{10} and V_{90} still remained significantly decreased post-48 compared to baseline values (Figures 3,4,5). MVIC decreased ($p < 0.01$) after DS, EO, and FW (Figure 6). Significant measurement x protocol interactions were detected for Dm ($F_{[5,59]} = 2.93$, $p = 0.020$) and V_{10} ($F_{[5,60]} = 3.12$, $p = 0.014$). Dm and V_{10} were significantly lower at post-train after protocols DS and FW compared to protocol PL ($p < 0.001$) (Figures 3 and 4).

Figure 3

Figure 4

Figure 5

Figure 6

Pearson correlation coefficients between mean protocols MVIC changes between baseline and post-train measurements and respective changes of Dm ($r = 0.64$; $p < 0.05$), and V_{10} ($r = 0.67$; $p < 0.01$) are presented in Figures 7 and 8. Besides, there was also a significant correlation between the same values of MVIC and V_{90} ($r = 0.66$; $p < 0.05$). No significant correlation ($p > 0.39$) was found between the skinfolds thickness from the thighs and TMG properties.

Figure 7

Figure 8

DISCUSSION

As far as we know, this is the first study to examine TMG muscle contractile properties after five different strength training protocols commonly used in the practical field. Whereas muscle force has been considered the best indicator of the ability of the muscle to perform (16), and fatigue studies are commonly based on measurements of muscle force (31), MVIC values were also measured after each protocol performance.

The lack of significant correlations between the skinfolds thickness from the thighs and TMG properties leads us to conclude that subcutaneous fat had no influence on these properties. Changes in neuromuscular function at post-train measured by TMG muscle properties Dm, V_{10} and V_{90} as well as a decrease in MVIC was observed after four (MS, DS, EO, FW) of the

five strength protocols. The reduction in Dm ($p < 0.001$) and in muscle contraction velocities V_{10} ($p < 0.01$), and V_{90} ($p < 0.01$) are in agreement with other studies which investigated the influence of strength exercises on TMG muscle properties (12,15). Moreover, the main reduction for these protocols in Dm (24.8%), V_{10} (18%), and V_{90} (17.2%) exceed the SEM values already reported above. Dm has been considered as a measure of muscle belly radial stiffness, and an increasing in such variable indicates smaller muscle belly radial stiffness, whereas its decreasing means greater muscle belly radial stiffness (12,15,25). An association between muscle fatigue and Dm decrease, as well as an increase in time of muscle contraction has been verified in biceps brachii muscle function one to six days after eccentric isokinetic contractions (15). These relationship could be explained by a reduced efficiency of the excitation-contraction coupling, impairment in membrane conducting properties and cellular structures destruction. This results in increased passive cellular structural tension, failure to fully activate the contractile machinery (lower number of cross-bridge bindings) and finally, in impaired muscle function (6,15,20,23,36). Altered intracellular Ca^{2+} regulation also likely played an important role in the observed reduced Dm ($p < 0.001$) and muscle velocities V_{10} ($p < 0.01$) and V_{90} ($p < 0.01$), as sarcoplasmic reticulum Ca^{2+} release rate has been markedly reduced after fatiguing muscle contractions (1,12). The significant Pearson correlation values ($r = 0.64\text{-}0.67$; $p < 0.05$) found in the present study are in accordance with Hunter al. (15), which demonstrate a positive correlation between the decrease in strength performance and changes in TMG muscle properties (Figures 7 and 8). As hypothesized, TMG was sensitive to changes in muscle force and different training determinants after lower-body strength training.

The changes in TMG muscle properties (Figures 3,4,5) and MVIC (Figure 6) point out that DS, EO and FW were the most stressful protocols in the present study. The higher stress signs evidenced by DS could be firstly explained because the number of repetitions at a high workload led to the highest TUT (28), compared to the other protocols (Table 1). Secondly,

according the size principle (26), during heavy strength exercise set, lower threshold motor units, composed of predominantly type I slow-twitch or type IIa fast-twitch muscle fibers are recruited first. As these motor units become fatigued, additional higher threshold motor units, composed of predominantly type IIx fast-twitch muscle fibers are recruited. However, the higher threshold motor units are fully recruited and subsequently fatigued only if muscle failure occurs, as in DS (37).

The protocols EO and FW may have a special effect on muscle performance and fatigue because of a greater eccentric muscle activation and subsequently higher exercise-induced muscle damage (21,27,32). Eccentric muscle actions have been shown to produce a greater amount of force than isometric or concentric actions despite a decreased motor units recruitment (32). The result is a higher tension produced per cross-bridge and progressive sarcomere overstretching, predisposing to destruction of contractile proteins and damage in cellular structures as sarcolemma, sarcoplasmic reticulum and t-tubules (24,32). The greater eccentric muscle activation in FW might be attributed to its inertial loading characteristics. According to Norrbrand et al. (22) and Tesch et al. (32), in traditional strength training devices, the maximal muscle activation occurs only at the very last concentric muscle repetitions, resulting in a failure to lift the particular load. Instead, flywheel resistance machines offer unrestrained resistance while the muscle shortens. Besides, eccentric forces greater than the concentric forces are possible. Indeed, a greater electromyographic activity during flywheel training was reported when compared with a traditional weight machine (21). Although electromyography measurement has not been performed in the present study, the higher levels of La after FW (Table 2) indicate also greater muscle activation and glycolytic demand, possibly from an increased recruitment of type II muscle fibers (14). The greatest post-exercise decrease in MVIC (Figure 6), the highest La after FW in post-train (Table 2), along with changes in TMG muscle properties (Figures 3,4,5) lead us to conclude a high influence of FW on neuromuscular disturbance.

In all training protocols examined, TMG muscle properties were statistically sensitive to the training load at post-train, with the exception of PL. As the TUT in PL was much lower comparing with the other protocols (see Table 1), the lack of statistical changes in TMG muscle properties in PL might be due to a very short ground contact time (18), a general advice in such exercise. We believe that the post-activation potentiation (PAP) also might play a role after PL execution. PAP is defined by enhanced muscle performance as a result of their contractile history (33) and it is believed that performance of a high-load exercise may excite the central nervous system, which enables a greater explosive performance in subsequent exercises. High-load lift may increase motor neuron excitability, motor unit recruitment patterns, or even increase activation of synergists, which allow an optimal training environment for explosive exercises, such as plyometric (19). However, in the present study, plyometric exercises were performed before the strength test (MVIC). Masamoto et al. (19) investigated the effects of different pre-event exercises, including a traditional warm-up protocol, tuck jumps or drop jumps, on 1RM squat performance in male athletes. An increase in 1RM squat exercise performance (3.5%) just after the execution of 2 DJ repetitions has been observed. This increase in 1RM squat performance was greater ($p < 0.05$) in comparison to traditional warm-up. The authors suggested that only plyometric exercises can raise neural stimulation to a level that will increase maximal muscle strength. Although muscle performance has not been enhanced in the present study after PL, it is possible that because the mechanisms associated to PAP, mainly the excitability of higher-threshold motor units (9,19), a reduction in muscle velocity or MVIC has been avoided.

Several TMG muscle properties clearly reflect the negative short term effect of most strength training protocols on muscle performance and neuromuscular function in the expected way (Figures 3,4,5,6). Nevertheless, the lower Tc values post-train ($p < 0.01$), which mean a shorter contraction time between 10 and 90% of Dm, apparently conduct in a paradox direction. On the other hand, in the light of the absolute decrease of Dm values, it became

comprehensive as post exercise T_c is related to a smaller displacement of the muscle belly. Accordingly, there was no correlation found between T_c and MVIC changes.

It can be concluded that some TMG muscle contractile properties are sensitive to changes in muscle force. D_m and contraction velocities V_{10} and V_{90} seem to be valid measures for neuromuscular assessment and might be used as fatigue indicator, while T_c has no relevance in this case. Training protocols that involve high number of repetitions, high and eccentric load and a longer TUT may induce higher changes in TMG muscle properties.

PRACTICAL APPLICATIONS

Performance tests have been considered the gold standard tests for fatigue diagnostic in sports practical field. However, such tests are physically demanding, generate fatigue and interfere in the training schedule. Our results demonstrated positive correlations between MVIC values and some TMG contractile properties, which indicate that such an analysis might be used as fatigue indicator. The absence of effects on muscle belly radial stiffness after PL indicates that such protocol is less demanding for the muscle tissue compared with the other protocols. The protocols characterized for a higher number of repetitions, load and time under tension provide higher fatigue levels. Ultimately, athletes and coaches should be cautious with intense or fatigue-inducing strength training, as it's necessary more than 48 hours for the TMG muscle properties recovery.

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FIGURE AND TABLE LEGENDS

Figure 1. Schematic illustration of study design (TMG = tensiomyography; 1RM = one repetition maximum; MVIC = maximal voluntary contraction; La = blood lactate; RPE = rate of perceived exertion; MS = Multiple Sets; DS = Drop Sets; EO = Eccentric Overload; FW = Flywheel; PL = Plyometrics).

Figure 2. TMG displacement sensor placed above rectus femoris.

Figure 3. Dm values through the different protocols (MS, DS, EO, FW and PL) at the measurement times baseline, post-train, post-24 and post-48. Values are mean $\pm$ SD. * $p < 0.01$: significantly different from baseline; # $p < 0.001$: significantly different to post-train; $^+p < 0.001$: significantly different to PL.

Figure 4. V_{10} values through the different protocols (MS, DS, EO, FW and PL) at the measurement times baseline, post-train, post-24 and post-48. Values are mean $\pm$ SD. * $p < 0.01$: significantly different from baseline; $^+p < 0.001$: significantly different to PL.

Figure 5. V_{90} values through the different protocols (MS, DS, EO, FW and PL) at the measurement times baseline, post-train, post-24 and post-48. Values are mean $\pm$ SD. * $p < 0.01$: significantly different from baseline.

Figure 6. MVIC values through the different protocols (MS, DS, EO, FW and PL) at the measurement times baseline and post-train. Values are mean $\pm$ SD. * $p < 0.01$: significantly different from baseline

Figure 7. Correlation for absolute changes of Dm (Δ Dm) and MVIC (Δ MVIC) between baseline and post-train measurements.

Figure 8. Correlation for absolute changes of V_{10} (Δ V_{10}) and MVIC (Δ MVIC) between baseline and post-train measurements.

Table 1. RM = repetition maximum; TUT = time under tension; ecc = eccentric phase; con = concentric phase; iso = isometric phase; acc = accelerating; x = intended explosive; btw = between; all-out = maximum effort.

Table 2. MS = Multiple Sets; DS = Drop Sets; EO = Eccentric Overload; FW = Flywheel; PL = Plyometrics. * $p < 0.05$: significantly different from all other protocols; # $p < 0.05$: significantly different to FW.

Table 1. Training protocols

Protocol	Multiple Sets	Drop Sets	Eccentric Overload	Flywheel	Plyometrics
Sets x reps	4 x 6	1 x 6 (+ 3 drop sets)	4 x 6	4 x 6 (+2 acc reps)	4 x 15
Intensity (% 1RM)	85	85 (70-55-40)	100 ecc/70 con	all-out	all-out
Repetition time (s)	2 ecc/x con	4 ecc/2 con	2 ecc/1 iso/x con	2 ecc/x con	maximum/explosive
Total TUT (s)	72	130-150	96	96	30
Rest btw sets (s)	180	-	180	180	5/180

RM = repetition maximum; TUT = time under tension; ecc = eccentric phase; con = concentric phase; iso = isometric phase; acc = accelerating; x = intended explosive; btw = between; all-out = maximum effort.

Table 2. Blood lactate concentrations and RPE values after the training protocols

Protocol	MS	DS	EO	FW	PL
La (mmol.L ⁻¹)	7.7 ± 2.8	8.6 ± 1.1	7.9 ± 2.6	12.0 ± 1.8*	3.0 ± 1.6*
RPE	8.6 ± 1.0 [#]	9.0 ± 0.9 [#]	9.2 ± 0.9	9.9 ± 0.3	5.2 ± 1.6*

MS = Multiple Sets; DS = Drop Sets; EO = Eccentric Overload; FW = Flywheel; PL = Plyometrics. *p < 0.01: significantly different from all other protocols; [#]p < 0.05: significantly different to FW.

weeks

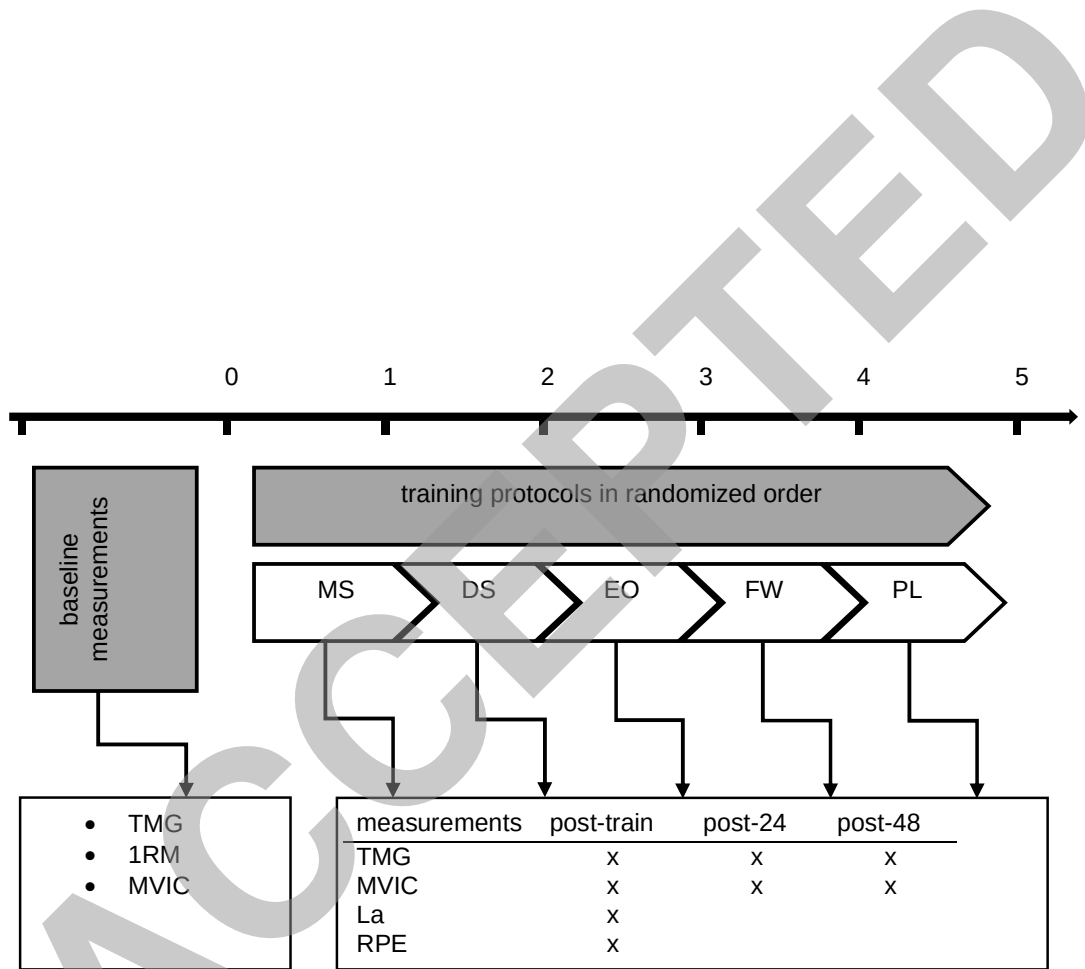


Figure 1. Schematic illustration of study design (TMG = tensiomyography; 1RM = one repetition maximum; MVIC = maximal voluntary contraction; La = blood lactate; RPE = rate of perceived exertion; MS = Multiple Sets; DS = Drop Sets; EO = Eccentric Overload; FW = Flywheel; PL = Plyometrics).



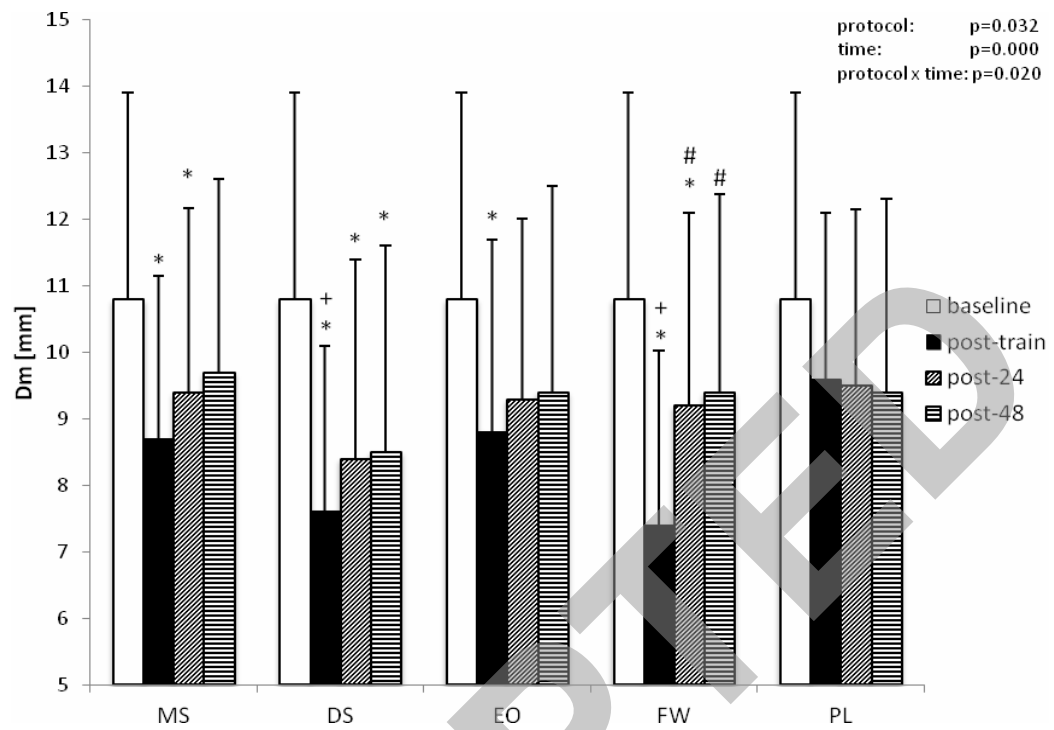


Figure 3. Dm values through the different protocols (MS, DS, EO, FW and PL) at the measurement times baseline, post-train, post-24 and post-48. Values are mean $\pm$ SD. * $p < 0.01$: significantly different from baseline; # $p < 0.001$: significantly different to post-train; + $p < 0.001$: significantly different to PL.

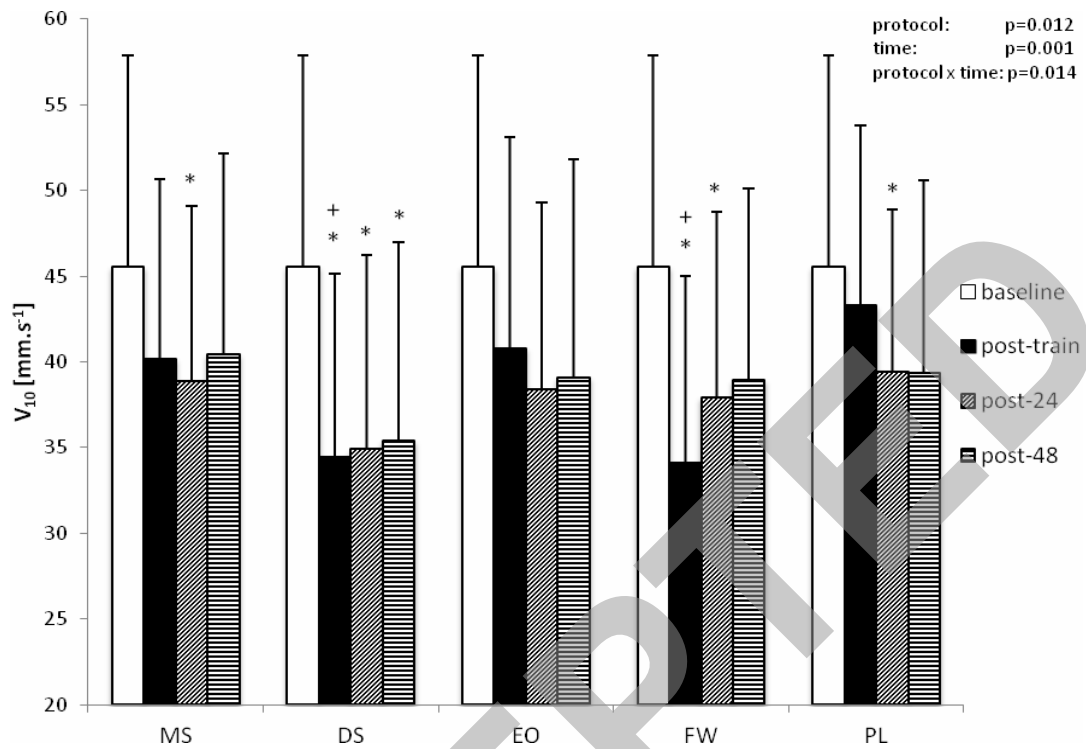


Figure 4. V_{10} values through the different protocols (MS, DS, EO, FW and PL) at the measurement times baseline, post-train, post-24 and post-48. Values are mean $\pm$ SD. * $p < 0.01$: significantly different from baseline; + $p < 0.001$: significantly different to PL.

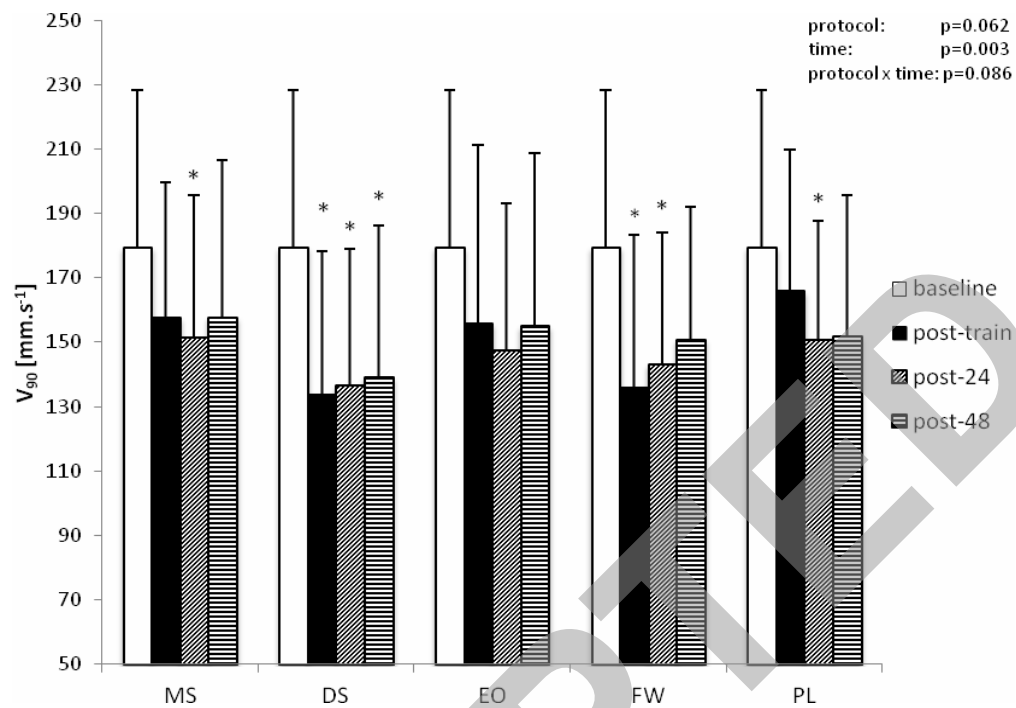


Figure 5. V_{90} values through the different protocols (MS, DS, EO, FW and PL) at the measurement times baseline, post-train, post-24 and post-48. Values are mean $\pm$ SD. * $p < 0.01$: significantly different from baseline.

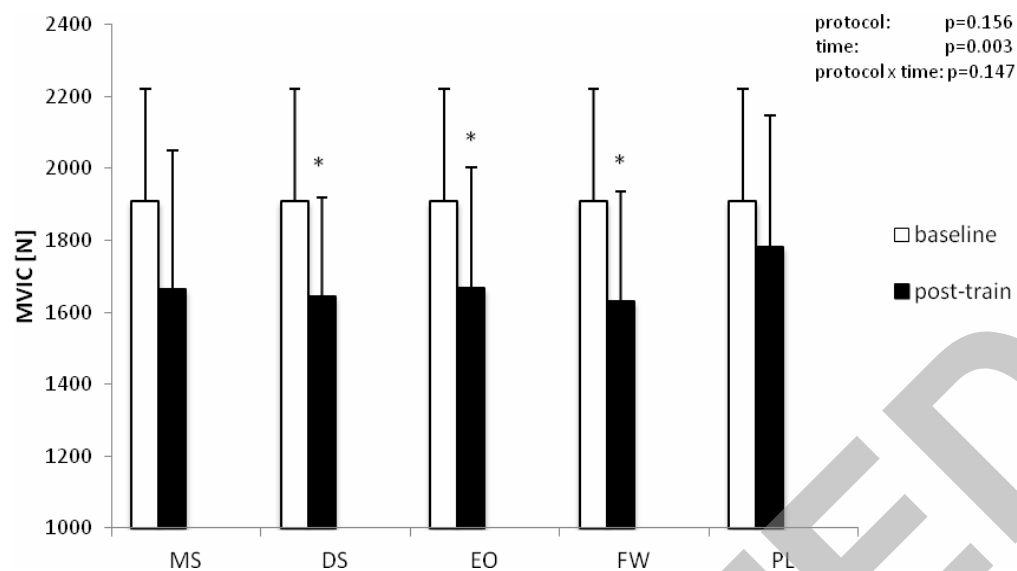


Figure 6. MVIC values through the different protocols (MS, DS, EO, FW and PL) at the measurement times baseline and post-train. Values are mean $\pm$ SD. * $p < 0.01$: significantly different from baseline

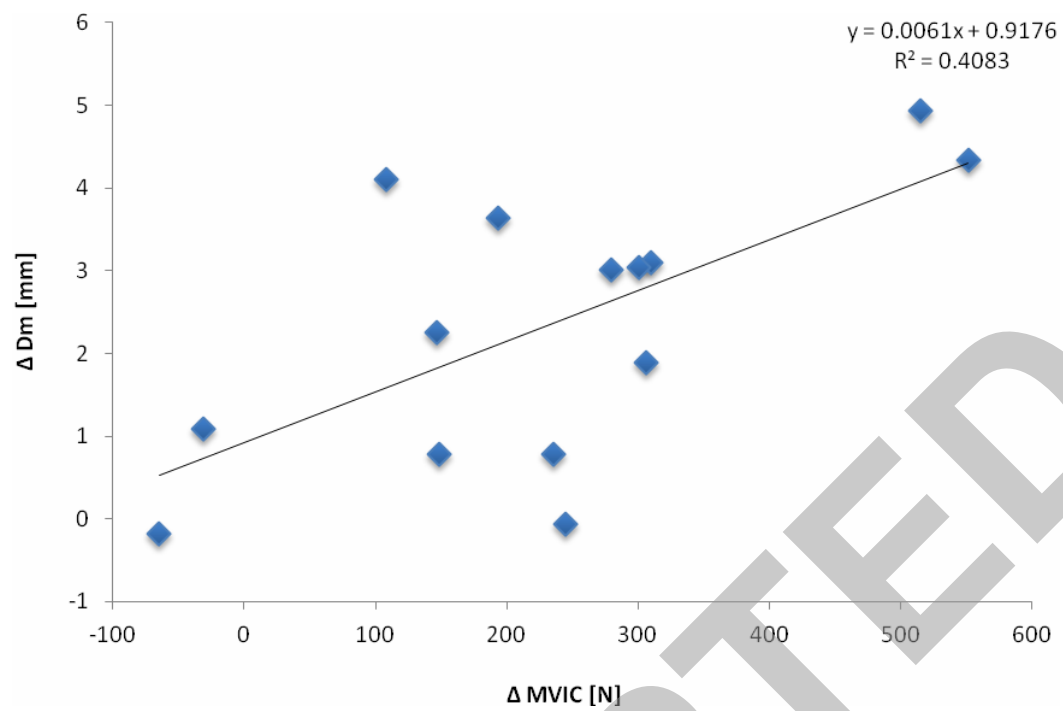


Figure 7. Correlation for absolute changes of Dm (Δ Dm) and MVIC (Δ MVIC) between baseline and post-train measurements.

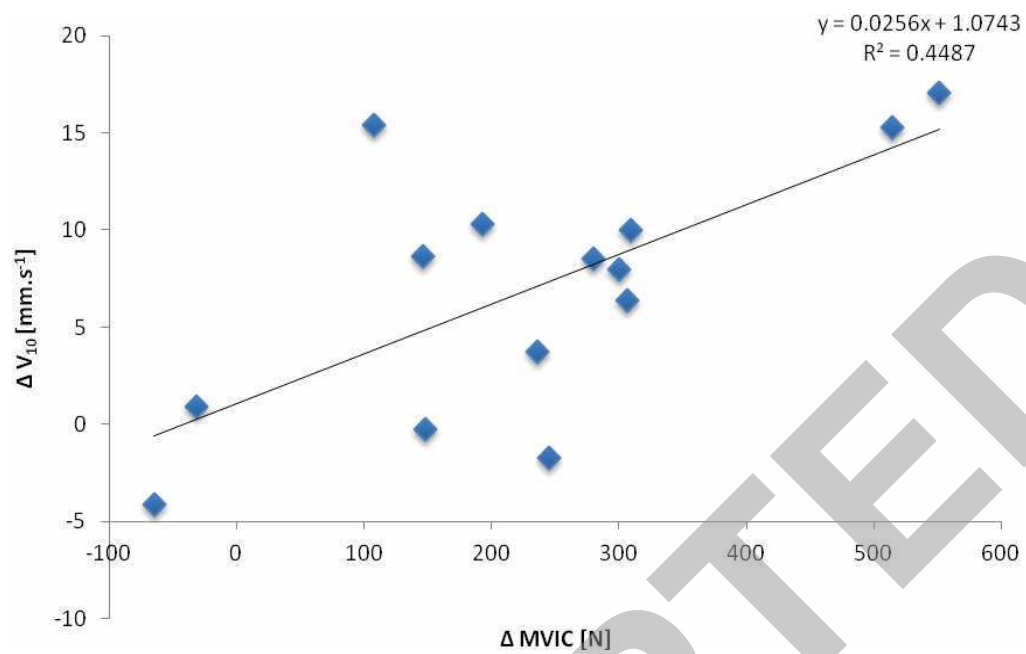


Figure 8. Correlation for absolute changes of V_{10} (ΔV_{10}) and MVIC ($\Delta MVIC$) between baseline and post-train measurements.